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COUNT: DS 1
RATE: DS 1
VALUE_1: DS 1
VALUE_2: DS 1
VALUE_3: DS 1
VALUE_4: DS 1
BAD: DS 1
TOGGLE: DS 1 ;Toggle every bit
ANS: DS 1 ;
ADDR: DS 1
STACK: DS 1
; ***MACRO'S***
SCL_HIGH MACRO
    SETB SCL ; SET SCL HIGH
    JNB SCL,$ ; LOOP UNTIL STRONG
                1 ON SCL
ENDM
; =====
CSEG ; Code begins here
; -----
; Main routine. Program execution
starts here.
; -----
ORG 00H ; Reset
ORG MAIN
ORG 000BH ;Timer Inter-
rupt0
JMP REFRESH
; -----
MAIN: MOV PSW,RR0 ; Select reg-
ister bank 0
MOV SP,#60H
MOV RATE,#00H
MOV COUNT,#00H
MOV VALUE_1,#01H
MOV VALUE_2,#02H
MOV VALUE_3,#03H
MOV VALUE_4,#04H
MOV alahr,#06H
MOV alamin,#30H
MOV TMOD,#01H ;enable timer0
for scanning
MOV TLO,#00H
MOV TH0,#0FFH
mov p3,0ffh ;make p3.2 as
timeset
clr p3.3 ;alarm off
CLR IT0
SETB ST0
SETB EA
;TIMER 0 SET
; global
;enable interrupts
SETB TR0 ;Start the Timer-0
MOV ANS,#00H ;clear temporary
toggle bit
CLR NEW
; =====
; INITILIZE RTC
; =====
SETB SDA ; ENSURE SDA HIGH
SCL_HIGH ; ENSURE SCL HIGH
CLR ACK ; CLEAR STATUS FLAGS
CLR BUS_FLT
CLR 2W_BUSY
CLR SQW
CALL OSC_CONTROL;Initilize the
RTC
ACALL SQW_CONTROL_1Hz
MOV keypr,#35H
; =====
; CHECK FOR ENTER THE TIME
CALL read_CLOCK
mov a,#06h
mov alahr,a
mov a,#30h
mov alamin,a
call store_clock
; =====
; MAIN PROGRAM
; =====
START_PROGRAM:
CALL READ_CLOCK
MOV A,HRS ;GET HRS, MIN
AND DISPLAY
anl a,#1fh
mov hrs,a
call alarm_chk
mov a,hrs
ANL A,#0F0H
SWAP A
MOV VALUE_1,A
MOV A,hrs
ANL A,#0F0H
MOV VALUE_2,A
MOV A,MINS ;GET HRS AND
DISPLAY
ANL A,#0F0H
SWAP A
MOV VALUE_3,A
MOV A,MINS
ANL A,#0F0H
MOV VALUE_4,A
CALL keycheck
AJMP START_PROGRAM
; =====
; alarm check
; =====
ALARM_CHK:
; COMPARES ALARMTIME WITH ACTUAL TIME-
AND ISSUES ALARM ON PIN11
MOV A,alahr
CJNE A,hrs,R ;HOURS COMPARE
MOV A,mins
CJNE A,alamin,R
;BOTH COMPARE EQUIL ISSUE ALARM FOR ONE
MINUTE
setb P3.3 ; A PULSE ON THIS PIN
TO SPEAKER via a
transfr
RET
clr p3.3
RET
; =====
DELAY: MOV R1,#0FFH
RET
; =====
REL: MOV R2,#0FFH
RE: NOP
DJNZ R2,RE
DJNZ R1,REL
RET
; =====
; One second delay routine
DELAYS: MOV R3,#0AH
RS3: MOV R1,#0FFH
RAL: MOV R2,#0FFH
RS2: MOV A,keypr
CJNE A,#35H,key ; if keypr is
; global not 35 then vary time
DJNZ R2,RS2
DJNZ R1,RAL
DJNZ R3,RS3
RET
; =====
keycheck:
key: jb timeset,nokey
mov a,mins
add a,#01
da a
mov mins,a
cjne a,#60h,storeclo
hrinc: mov a,hrs
add a,#01
da a
mov hrs,a
cjne a,#13h,storeclo
windhour: mov hrs,#1
storeclo: call store_clock
jmp nokey
ala: mov a,mins
mov alamin,a
mov a,hrs
mov alahr,a
call store_clock
jmp kk
nokey: jb p3.3,ala
kk: MOV R1,#100
call dela ;TO make numbers move
slowly for adjusting
RET
; =====
; 7-SEGMENT DISPLAY ROUTINE
; =====

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